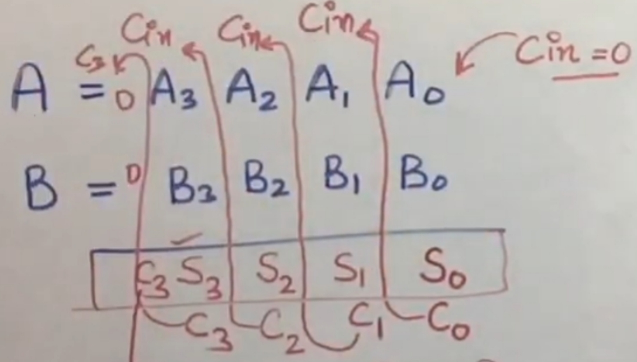
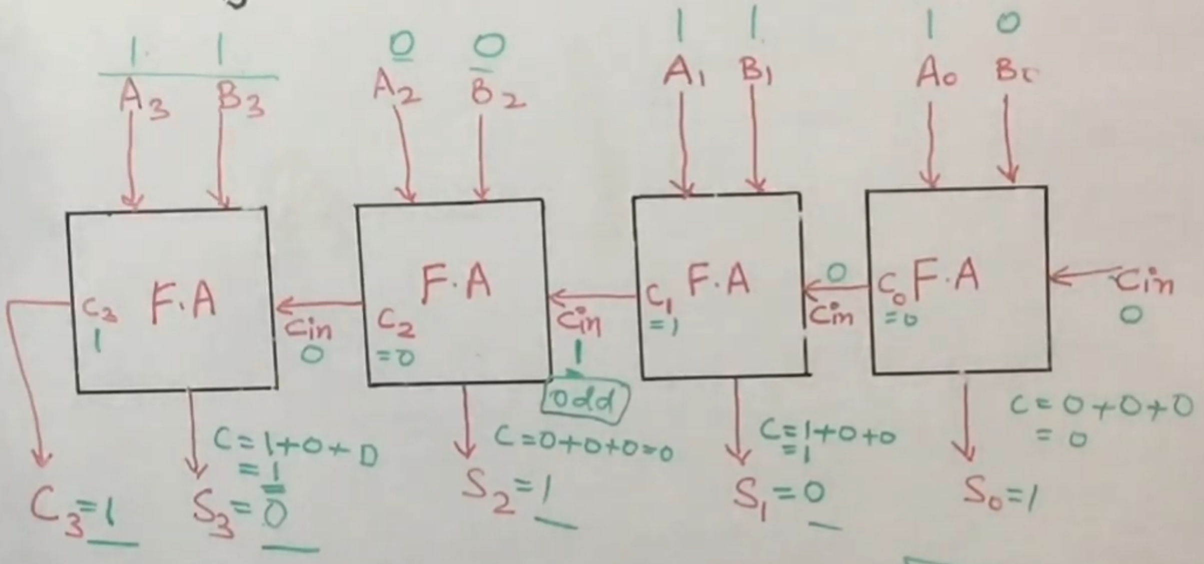
Four Bit Parallel Adder using Full Adder



$\boxed{C_3 = 0 \text{ or } 1}$ {4 bit, 5 bit}

$$
\begin{array}{r}
 A_3A_2A_1A_0 \\
A = 1011 = 11 \\
B = 1010 = 10 \\
 B_3B_2B_1B_0 \\
\hline
10101 = 21 \\
16\ 8\ 4\ 2\ 1
\end{array}
$$

$\boxed{10101}$



$S = A \oplus B \oplus C$

$C = AB + BC_{in} + AC_{in}$

Ex-OR
odd 1s detector
High

IC 74HC283

A = 1011 = 11

B = 1010 = 10

$B_3\ B_2\ B_1\ B_0$

1 0 1 0 1 = 21 ✓

16 8 4 2 1